

* DMR *

Q.1 write a short note on digital twin.

⇒ A digital twin is a dynamic, virtual replica of a physical object, process or system that uses real time data from IOT sensors.

Types

- ① Component twins (CDT)
- ② Asset twins (ADT)
- ③ system twins (SDT)
- ④ Process twins (PDT):

Key Components

- ① Physical Asset: The real world object with IOT sensors.
- ② Virtual model: A digital replica updated with sensor data (IOT)
- ③ Data pipeline: The connection that synchronizes the physical asset with its virtual counterpart in real-time.
- ④ Analytics & AI: machine learning and AI are often applied to the data to detect patterns, simulate future scenarios & enable predictive maintenance

Applications

- Predictive maintenance
- manufacturing optimization
- Product development & design
- health care.

Q2 Explain Robots and effectors

- Robot
- A robot is an programmed machine designed to sense its environment, process information and execute physical tasks automatically.
- They enhance efficiency by performing repetitive, precise or dangerous work.
- Applications

① Manufacturing: Industrial robot arms for assembly, painting and welding.

② Household: Autonomous vacuums like Roombas.

③ Medical/Healthcare: Robotic surgical assistants

④ Exploration: Space rovers on other planet.

• Effectors

- End effectors are peripheral devices attached to the end of a robotic arm, allowing it to interact with its environment to perform specific tasks.

• Types

① Tools: Devices designed to perform work on a piece, such as spray guns, welders, drills

② Grippers: Devices that hold, or grab parts, such as vacuum cups, magnetic grippers or mechanical fingers.

• Applications

① Pick and place

② Assembly

③ Material handling & processing

④ Surface treatment.

Q3 Describe Automatic Tool changer (ATC)

⇒ An Automatic Tool changer (ATC) is a CNC machine component that automatically exchanges cutting tools in the spindle.

- It consists of a tool magazine (storage) and a gripper mechanism, enabling rapid, precise switching between tools for complex, high-production machining, such as milling/turning.

• Types

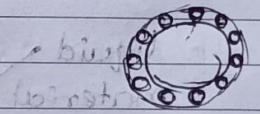
① Drum

- A horizontal or vertical rotating disc.

- used on smaller machines

② Chain-type

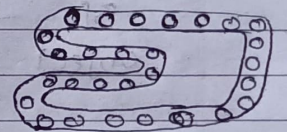
- Tools are arranged on a chain



③ Rack/Linear

- A rack of tools, common on

CNC routers.



• Advantages

⇒ Increases production output.

⇒ Improves safety.

⇒ Improve precision & consistency.

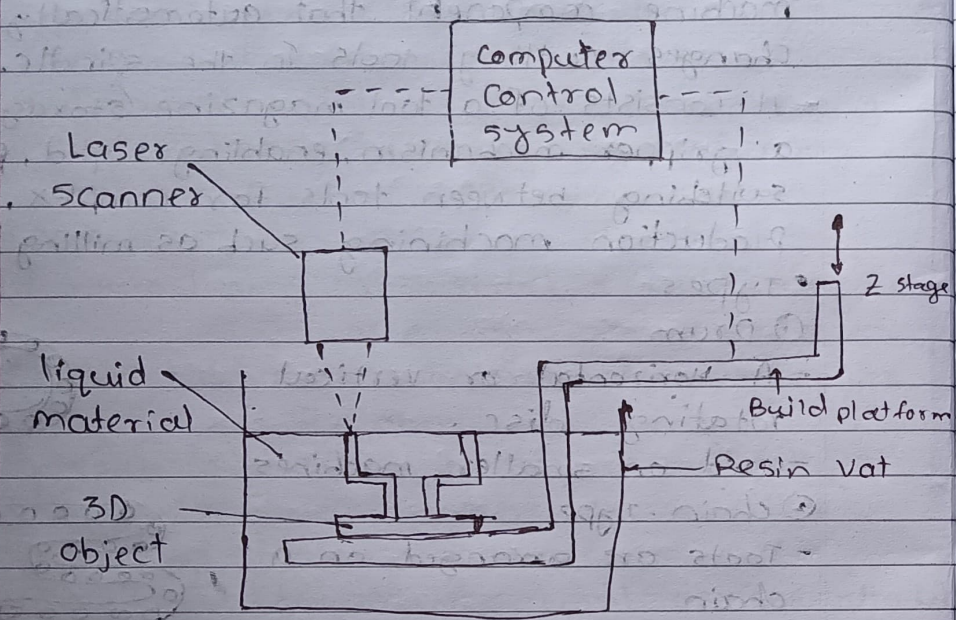
⇒ Reduce labour & operating costs.

Q4 What is stereo lithography apparatus (SLAs)

⇒ Stereolithography Apparatus (SLA) is an Additive manufacturing technology that creates high-accuracy, solid plastic 3D objects from liquid polymer resin using photo

a UV laser

Construction



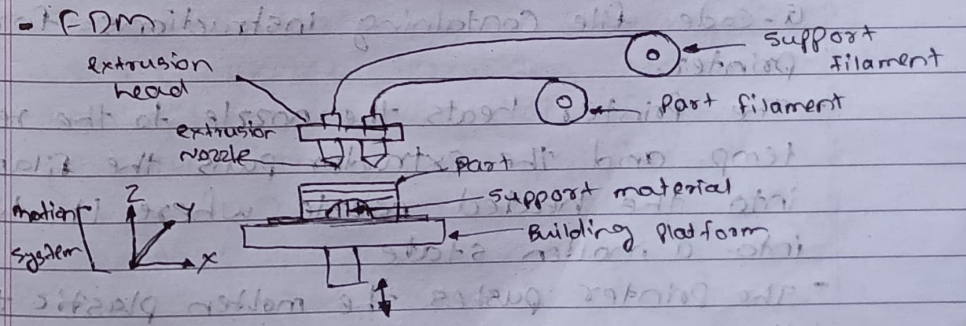
- Resin vat : A transparent container filled with liquid.
- UV light source : A laser beam is used in (Laser) traditional SLA.
- Build platform: A platform attached to the elevator that lowers into the vat to manage layer thickness & moves up & down.
- Computer : Processes the CAD file and computer control system controls the laser scanning.
- Working
 - A 3D model is created on a computer and sliced into hundreds or thousands of thin 2D layers.
 - A build platform lowers into the vat

vat of liquid resin, leaving a very thin gap at bottom.

- A UV laser acts like a digital pen, drawing the shape of the first layer onto the liquid resin.
- The liquid turns solid wherever the light touches.
- The platform raises slightly, allowing fresh liquid resin to flow underneath the hardened layer.
- Repeating this entire process until the entire 3D object is formed.
- After forming a 3D object, it is removed washed in alcohol to remove extra resin.

Q.5 What is additive manufacturing process. Explain FDM technique with application.

⇒ Additive manufacturing is a computer-controlled process that creates 3D objects by depositing material layer-by-layer based on digital data, typically using a CAD model.



- Fused Deposition Modeling (FDM) is an additive manufacturing technique that builds part layer-by-layer by extruding thermoplastic filaments through a heated nozzle.

• Construction

① Filament spool: The thermoplastic material (eg. PLA, ABS) is used.

② Extrusion head: A heated chamber that melts the plastic to a precise temp.

③ Nozzle: The tip attached to the hot end, Common sizes range from 0.2 mm to 0.8 mm.

④ Build platform: The base where object is printed.

⑤ Motion system: motors and belts that move (X, Y, Z) the print head horizontally and vertically to deposit material along the programmed path.

• Working

- A CAD model is sliced by software into thin horizontal layers and converted into a G-code file containing instructions for the printer.

- The printer heats the nozzle to the required temp and the extruder pulls the filament into the heated nozzle, where it turns into a molten state.

- The printer pushes the molten plastic through the nozzle, which moves to draw the 2D

cross-section of the first layer onto the build platform.

- As soon as the material is extruded, it cools down and solidifies adhering to the layer below it.

- Once a layer is finished the nozzle moves up to start the next layer.

• Advantage

- cost effective

- material versatility

- Rapid prototyping

- Easy customization

• Disadvantage

- surface finish is rough

- Interlayer bonding

- is weather

- Dimensional accuracy

- Prone to warping.

• Applications

- Jigs & fixture

- manufacturing customized parts such as covers, brackets

- Building anatomical models, teaching aids.

- Building engineering prototypes.

Q6 Explain Computer Aided Process planning (CAPP)

⇒ Computer-Aided process planning (CAPP)

uses computer technology to automate and optimize the creation of manufacturing processes, bridging the gap between product design (CAD) and manufacturing (CAM).

• Advantage

① Improved productivity & speed

- reduces the time required to generate

Process Plans

- ③ Consistency and quality
 - produces similar parts using similar processes
- ③ Reduce Cost
 - lower production cost
- ④ Flexibility and optimization
 - Allows for faster iteration for new designs and easy modifications.

Q.7 Describe agile manufacturing.

⇒ Agile manufacturing is a methodology focused on speed, flexibility and rapid response to changing customer demands and market shifts.

Key characteristics

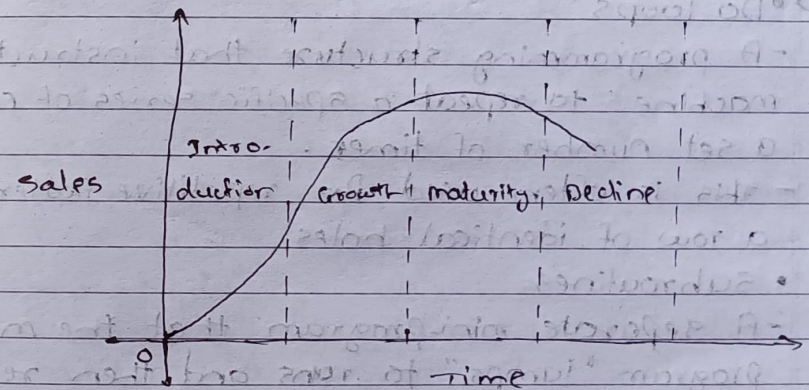
- ① Rapid responsiveness
 - Adapting to sudden changes in market demand or customer preferences.
- ② Customization
 - meeting individual customer needs rather than just mass production.
- ③ Technology integration
 - using robotics, IoT and AI to streamline workflows and increase flexibility.
- ④ Knowledge-Driven
 - Empowering the workforce with data, analytics and information to make fast decisions.
- ⑤ Partnerships
 - Utilizing collaboration with suppliers and partners to enhance capabilities.

Advantages

- Faster Time-to-market
- Increased Competitiveness
- Improved Customer Satisfaction
- Reduced Risk.

Q.8 What is the product life cycle.

⇒ The product life cycle (PLC) is the series of stages a product goes through, from its initial development to its final removal from the market.



PLC Phases

- ① Introduction stage
 - The product is launched into the market. Sales are low and costs are high due to heavy promotion and R&D.
- ② Growth stage
 - The product gains popularity, leading to a rapid increase in sales and subsequently profits.

③ Maturity stage

- sales growth becomes stable, as the product is widely accepted, making this often the most profitable stage.

④ Decline stage

- sales and profits fall due to market saturation, changes in consumer taste or superior competitors.

Q.9 Explain Do loops, Canned cycles, Subroutine.

Explain how canned cycles are used in writing program write G code used for canned cycles.

⇒ • Do loops

- A programming structure that instructs the machine to repeat a specific series of commands a set number of times.

- This is often used for repetitive patterns, like a row of identical holes.

• Subroutines

- A separate mini-program that the main program "jumps" to runs and then returns from.

- Use these for complex custom shapes (you need to repeat in different locations).

• Canned cycle

- A canned cycle is a preprogrammed sequence of movements built into the machine's controller.
- One single line of G-code can replace dozens of individual movement commands.

• How canned cycles are used

① Define the cycle

- You select the type and provide parameters like depth and feed rate.
- ex. G81 Z-0.5 R0.1 F5.0

② Position the tool

- You list the X and Y coordinates for each spot where you want the cycle to run.

③ Cancel the cycle

- You must use G80 to tell the machine the repetitive operation is finished.

• Common canned cycle G-code

G81 - simple drilling

G82 - spot drilling

G83 - Peck drilling

G84 - Tapping

G80 - Cancel cycles

• G-code example

G81 Z-0.5 R0.1 F5.0

X1.0 Y1.0

X2.0

X3.0

G80

Q.10 What do you mean by Industry 4.0? How it concern with manufacturing.

⇒ Industry 4.0 is the current trend of automating manufacturing using smart technology, data and internet connectivity.

- It connects machines to computers, allowing factories to analyze data, make autonomous decisions, and optimize production.

• How industry 4.0 affects manufacturing.

① Smart machines (IIOT)

- Factory machines are connected to the internet and each other, sharing data in real-time to work together efficiently.

② Predictive maintenance

- machines predict when they will break down, allowing repairs before a shutdown happens, reducing downtime.

③ Mass customization

- Systems are flexible enough to efficiently produce personalized products, allowing for custom orders at mass-production speeds.

④ Data driven decisions

- AI analyzes production data to identify defects, minimize waste and improve quality.

⑤ Automated processes

- Advanced robotics and systems automate complex task, reducing manual labor and increasing safety.

Q.11 short note on maintenance of CNC machines

⇒ CNC machine maintenance is the proactive care and upkeep required to preserve the accuracy, functionality, and reliability of high precision machine tools.

- A structured maintenance program is typically

divided into specific intervals, to manage different systems logically.

① Daily maintenance

- Cleanliness: removes metal chips, dust and debris from work areas.
- Fluid level: check and replace coolant, hydraulic oil and lubrication levels.

- Visual & audio: Listen for unusual noises and inspection. Check for leaks.

② Weekly maintenance

- Cooling system: inspect and clean cooling fans and filters to prevent overheating.

- Machine components: visually inspect belts and pulleys for wear.

- Tool changer: Lubricate the tool changer and ensure tool heads are held securely.

③ Monthly & Quarterly maintenance

- Deep cleaning: Clean chip collectors and radiator grills.

- Alignment: Inspect and adjust jaws, chucks and fixtures for proper alignment.

- Greasing: Apply grease to chains, conveyors and linear guides.

④ Annual maintenance:

- Calibration: Perform full axis calibration and check for backlash in tool heads.

- Fluid replacement: Drain, clean and refill all lubricant and hydraulic tanks.

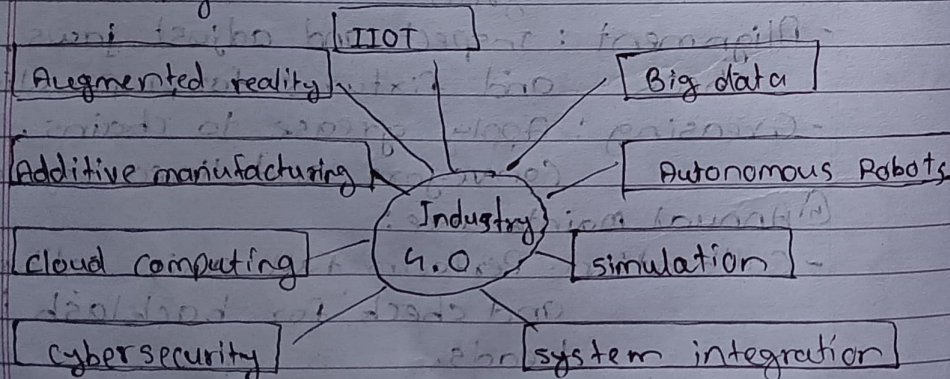
- Software updates: Install the latest firmware to improve performance and security.

Q.12 Draw and explain Industry 4.0 wheel and also describe cyber-physical system.

⇒ • Cyber physical system (CPS)

- A cyber-physical system is the foundational architecture of industry 4.0.
- It is a mechanism where physical objects are integrated with computational algorithms and networked communication.
- The physical part involves the actual hardware, sensors and actuators that perform tasks in the real world.
- The cyber part involves the software, data models and communication networks that monitor and control the physical entities.

• Industry 4.0



① Industrial Internet of Things (IIOT)

- Sensors placed on machines that collect real-time data about temperature, speed and health.
- Connecting physical machines to the digital world.

② Big data

- The vast amount of information collected by sensors is analyzed to make faster, better decisions and identify inefficiencies.

③ Cloud Computing

- Data is stored and processed on remote servers allowing manufacturers to access info from anywhere in the world.

④ Autonomous Robots

- Advanced AI-powered robots that can perform complex tasks, collaborate with humans and adapt to changes in production.

⑤ Additive manufacturing (3D Printing)

- Creating complex customized parts layer-by-layer, reducing material waste and enabling on demand production.

⑥ simulation (digital twin)

- Create a virtual replica of physical machine to test and optimize performance before making changes in real life.

⑦ Cybersecurity

- Protecting these connected systems from hackers and cyberattacks.

⑧ Augmented Reality (AR)

- Using digital displays to assist workers in real-time, such as showing repair instructions directly on machine.

Q13 Write a short note on IOT-enabled manufacturing system.

⇒ An IOT-enabled manufacturing system (often called IIOT) is a modern factory setup where machines, sensors and computers are connected to the internet to share data in real-time.

• Working

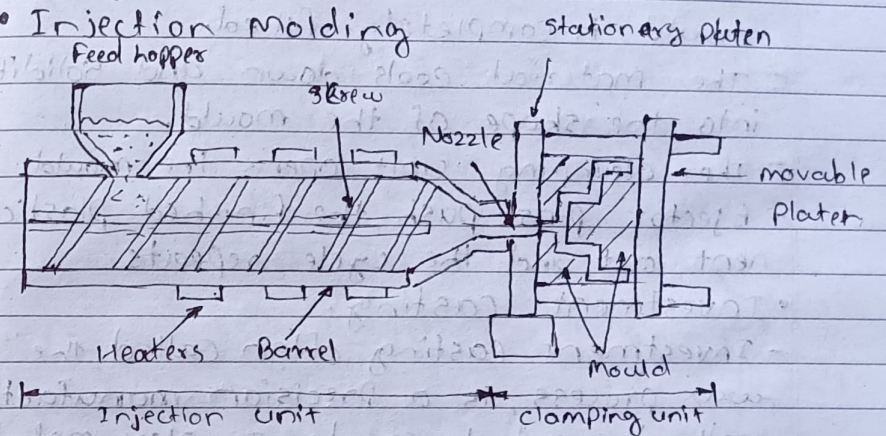
- ① Sensors : Gather data (temp, speed, vibration) from machinery.
- ② Connectivity : Transmits this data via Wi-Fi, 5G or Ethernet to cloud.
- ③ Data Analysis : AI and software analyze the data to spot patterns or problems.
- ④ Action : The system automatically adjusts machinery or alerts a human operator to take action.

• Benefits

- Predictive maintenance
- Real-time monitoring
- Better quality control
- Energy efficiency

Q14 short note on injection molding and investment casting.

⇒ • Injection Molding



• Construction

- Hopper : A container where raw plastic pellets are poured in.
- Barrel : A heated cylinder containing a reciprocating screw.
- screw : Pushes plastic forward, using friction and heat to melt it.
- Nozzle : The tip where molten plastic leaves the barrel to enter the mold.
- Mould : A two-part metal cavity shaped to create the desired product.
- clamping unit : A powerful system that holds the mould halves together against high injection pressure.

• Working

- The two halves of the mould are pressed together strongly by the clamping unit.
- molten plastic is forced through the

nozzle into the closed mould cavity.

- Pressure is maintained to ensure the mould is completely filled.
- The material cools down and solidifies into the shape of the mould.
- The clamping unit opens the mould halves.
- Ejector pins push the finished plastic component out and the cycle repeats.

• Investment Casting

- Investment Casting, often called the "lost wax process", is a precision manufacturing method used to create complex metal parts.
- Steps of investment casting

① make a wax pattern

- A wax replica of the final part is created, usually by injecting wax into a metal die.

② Assemble the "Tree"

- Multiple wax patterns are attached to a central wax stick forming a "tree" structure.

③ Build the shell

- The wax tree is dipped into a fine liquid ceramic slurry and then coated with sand. This is repeated several times to create a strong, thick ceramic shell.

④ Remove the wax (Lost wax)

- The ceramic shell is placed in an oven or autoclave. The wax melts and runs out, leaving a hollow ceramic mold.

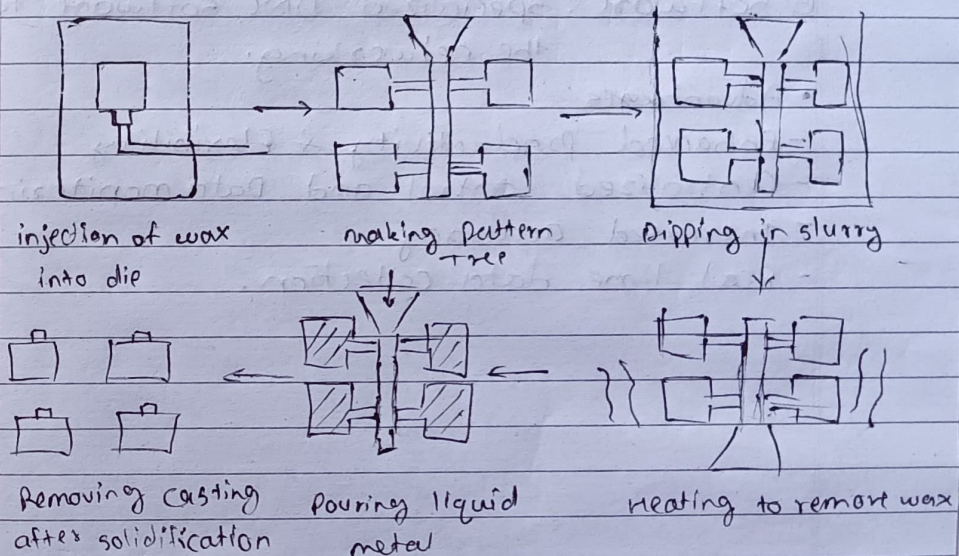
⑤ Pour the metal

- The empty ceramic mold is heated to

a high temperature and molten metal is poured into it.

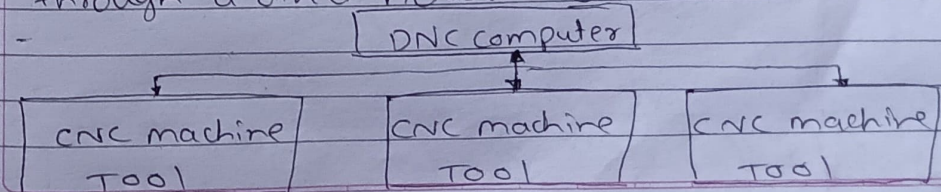
⑥ Finalize the part

- Once the metal cools and solidifies, the outer ceramic shell is broken away and the metal parts are cut from the tree and finished.



Q15 Short Note on direct Numerical control (DNC)

→ Direct Numerical Control (DNC) is a system where a single central computer controls multiple CNC machines at the same time through a direct network connection.



• Key Components of a DNC system

- ① Central Computer : stores and manages part programs.
- ② Communication : cables connecting the computer network to the machines.
- ③ Machine Tools : The CNC machines (mills, lathe) receiving the data.
- ④ Software : specialized DNC software to handle the networking.

• Advantages

- Enhanced productivity & Flexibility
- Centralized Control and Data monitoring
- Increased accuracy
- Real time data collection.

